CAMPUS PEER-TO-PEER MARKETPLACE ("CAMPUSBAZAAR")

A Specialized Institutional Resource Sharing & Micro-Rental Platform for College Students

A Capstone Project Report Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY / BACHELOR OF ENGINEERING IN COMPUTER SCIENCE & ENGINEERING

Course: Object-Oriented Programming Using Java (Core Java, MVC Architecture & MySQL)

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UNDER THE SUPERVISION OF:
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Department of Computer Science & Engineering
Academic Year: 2025 – 2026

CERTIFICATE OF PROJECT COMPLETION

This is to certify that the project entitled **"CAMPUS PEER-TO-PEER MARKETPLACE (CampusBazaar)"** is a bonafide work carried out by the student team in partial fulfillment of the requirements for the academic evaluation of Object-Oriented Programming Using Java in the Department of Computer Science and Engineering during the academic session 2025–2026.

The project demonstrates sound software engineering principles, rigorous application of Core Java Object-Oriented Programming paradigms (Encapsulation, Inheritance, Polymorphism, Abstraction, and Interface Segregation), robust relational database modeling in MySQL using JDBC PreparedStatement, an MVC architectural separation, and an automated verification suite covering unit, integration, and algorithmic correctness.

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Internal Project Guide
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External Examiner 1
Date: November 12, 2026

External Examiner 2
Institutional Seal

EXECUTIVE ABSTRACT

Within university campuses worldwide, tertiary students face severe cyclical financial outlays at the beginning of each academic semester. Expensive textbooks, scientific calculators, laboratory drawing tools (mini-drafters), hostel appliances, and bicycles represent high initial investments. Once a semester concludes, these valuable assets sit idle in dormitory rooms, while incoming junior cohorts purchase identical items from off-campus retail vendors at full price. Existing informal channels (e.g., WhatsApp and Telegram groups) lack trust, discoverability, rental tracking, and payment accountability, while generic classifieds (e.g., OLX, Quikr) fail due to lack of student verification, stranger danger, and logistical distance.

To solve this problem, we present CampusBazaar — an institutional, closed-loop peer-to-peer e-commerce and micro-rental platform engineered exclusively for university students and campus personnel. Developed in Core Java (JDK 17+) strictly adhering to an academic Model-View-Controller (MVC) architecture, the system incorporates: (1) Zero-Trust Student Identity Verification; (2) Polymorphic Resource Listings supporting outright peer sales, daily micro-rentals, and direct barter swaps; (3) An Algorithmic Rental Engine that calculates exact day durations and late-return penalty deductions from security deposits; (4) A Rule-Based Smart Pricing Engine that evaluates category depreciation and condition retention to prevent price gouging; (5) Safe Campus Landmark Handovers verified via cryptographic 6-digit OTP PINs; (6) Multi-Round Counter-Offer Price Bargaining; (7) Official Digital Clearance Receipts carrying SHA-256 integrity verification hashes for administrative hostel clearance; and (8) A Dual Presentation Layer comprising an interactive Java Swing Desktop GUI and an embedded zero-dependency Web Portal (http://localhost:8080) featuring real-time aesthetic theme customization across 8 harmonious color palettes.

The system's persistence layer features a production-ready 15-table relational schema in MySQL 8.0 integrated via JDBC PreparedStatement (ensuring SQL injection immunity), alongside a thread-safe in-memory Collections DataStore enabling instant zero-setup evaluation. The implementation is verified by an automated test suite comprising 37 comprehensive test cases achieving a 100% pass rate (37 PASSED | 0 FAILED). The resulting platform delivers a secure, sustainable circular economy for campuses, significantly lowering educational costs while demonstrating exemplary Java OOP engineering rigor.

CHAPTER 1: INTRODUCTION

1.1 Project Overview

CampusBazaar is an institutional peer-to-peer commerce and resource-sharing platform engineered in Core Java. It establishes a trusted digital circular economy exclusively for university students and campus residents.

1.2 The Campus Dilemma

Students face recurring, sharp semester expenses for academic textbooks, calculators, mini-drafters, and cycles. These items become dormant post-exams while junior cohorts repurchase them at retail prices.

1.3 Academic Objectives

• Master Java Object-Oriented Programming (Inheritance, Polymorphism, Abstraction, Encapsulation, Interface Segregation).
• Implement an MVC + DAO architectural pattern with zero external framework overkill.
• Build mathematical engines for micro-rental penalty settlement and rule-based smart pricing.
• Achieve 100% automated test coverage verified across 37 test cases.

CHAPTER 2: EXISTING SYSTEM VS PROPOSED SYSTEM

Comparative Analysis Matrix

Parameter	Informal (WhatsApp)	Generic (OLX)	CampusBazaar
Verification	None	Open Public	Institutional Roll Verified
Discoverability	Unstructured Chat	Metro-Level Search	Structured Campus Catalog
Rental Engine	None / Friction	Not Supported	Algorithmic Late Penalty Math
Handovers	Unregulated	Risky Courier/Meet	Designated Spots + 6-Digit OTP
Receipts	None	Basic Invoice	SHA-256 Clearance Vouchers
Architecture	Black-box	Closed Web	Core Java OOP (MVC + DAO)

CHAPTER 6: OBJECT-ORIENTED PROGRAMMING IMPLEMENTATION

CampusBazaar demonstrates concrete Core Java OOP paradigms:
• **Encapsulation:** Private fields with validated getters/setters preventing negative prices and invalid rental bounds.
• **Inheritance:** Abstract class Listing specialized by SaleListing, RentalListing, and BarterListing.
• **Polymorphism:** Dynamic method dispatch via getEffectivePrice() resolved at runtime across heterogeneous collections.
• **Interface Segregation:** Decoupled behavioral contracts: Searchable, Rentable, Reviewable, Reportable.
• **Custom Exceptions:** CampusMarketplaceException, ValidationException, InvalidRentalDurationException.
• **Dual Persistence:** Production MySQL 8.0 with automated fallback to in-memory concurrent DataStore.

CHAPTER 9: AUTOMATED TEST VERIFICATION (37 TESTS)

Verification Result: ■ 37 PASSED | 0 FAILED (100% Success Rate)

Test ID	Module / Scenario	Expected Output	Status
TC-01.1	SaleListing.getEffectivePrice()	Resolves selling price 450.0	**PASS**
TC-01.2	RentalListing.getEffectivePrice()	Resolves daily rate 35.0	**PASS**
TC-01.3	Dynamic Polymorphic Dispatch	Resolved via abstract Listing ref	**PASS**
TC-02.1	Rental standard charge (5 days @ 40)	Total charge = ■200.0	**PASS**
TC-02.2	Rental late penalty (3 days overdue @ 25)	Late fee ■75.0 deducted from deposit	**PASS**
TC-03.1	Smart Pricing target estimation	Target price ~ ■510.0	**PASS**
TC-04.1	SHA-256 Hash Length check	Exactly 64 hexadecimal characters	**PASS**
TC-05.1	Invalid email syntax check	Throws ValidationException	**PASS**
TC-05.3	Inverted rental date bounds check	Throws InvalidRentalDurationException	**PASS**
TC-08.2	Safe Handover OTP PIN creation	Generates 6-digit numeric PIN	**PASS**
TC-08.4	Correct Handover PIN verification	Handover status = VERIFIED_COMPLETED	**PASS**
TC-10.2	Clearance slip digital signature	SHA-256 hash generated (>= 16 chars)	**PASS**

Full 37/37 automated test suite logs and details available in docx report and docs/MILESTONE_10_TEST_REPORT.md.